.Create Three Java Classes into the project. Name them WCDriver(having the main function), WCMapper, WCReducer.

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase implements
Mapper<LongWritable,Text, Text, IntWritable> {

	// Map function
	public void map(LongWritable key, Text value, OutputCollector<Text,
				IntWritable> output, Reporter rep) throws IOException
	{

		String line = value.toString();

		// Splitting the line on spaces
		for (String word : line.split(" "))
		{
			if (word.length() > 0)
			{
				output.collect(new Text(word), new IntWritable(1));
			}
		}
	}
}
```

Reducer Code: You have to copy paste this program into the WCReducer Java Class file.

```
 // Importing libraries
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements
Reducer<Text,IntWritable, Text, IntWritable> {

	// Reduce function
	public void reduce(Text key, Iterator<IntWritable> value,
				OutputCollector<Text, IntWritable> output,
						Reporter rep) throws IOException
	{

		int count = 0;

		// Counting the frequency of each words
		while (value.hasNext())
		{
			IntWritable i = value.next();
			count += i.get();
		}

		output.collect(key, new IntWritable(count));
	}
}
```

Driver Code: You have to copy paste this program into the WCDriver Java Class file.

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

	public int run(String args[]) throws IOException
	{
		if (args.length < 2)
		{
			System.out.println("Please give valid inputs");
			return -1;
		}

		JobConf conf = new JobConf(WCDriver.class);
		FileInputFormat.setInputPaths(conf, new Path(args[0]));
		FileOutputFormat.setOutputPath(conf, new Path(args[1]));
		conf.setMapperClass(WCMapper.class);
		conf.setReducerClass(WCReducer.class);
		conf.setMapOutputKeyClass(Text.class);
		conf.setMapOutputValueClass(IntWritable.class);
		conf.setOutputKeyClass(Text.class);
		conf.setOutputValueClass(IntWritable.class);
		JobClient.runJob(conf);
		return 0;
	}





// Main Method
	public static void main(String args[]) throws Exception
```

```
	{
		int exitCode = ToolRunner.run(new WCDriver(), args);
		System.out.println(exitCode);
	}
}
```

3. create a file on Desktop(sample.txt) and type the below lines:

```
hi how are you
how is your job
how is your family
how is your brother
how is your sister
```

```
save your file in  /home/hadoop/input
```

```
5. View the directory content
hadoop fs -ls /
```

```
6.  Create a directory using the following command. If any directory existing,
use the same directory for the command

hadoop fs -mkdir /inputbda
```

7. Copy the file into HDFS

```
hadoop fs -copyFromLocal  /input/sample.txt  /inputbda  or

hadoop fs -put  /input/sample.txt  /inputbda

hadoop fs -ls /inputbda
```

```
8.Run the Map Reduce Program
hadoop jar /home/hduser/Desktop/wordcount.jar WCDriver /inputbda/sample.txt
/outputbda

9. hadoop fs -ls /outputbda

10. hadoop fs -cat /outputbda/part-00000
```

OUTPUT:

```
root@hp:~# mkdir ~/wordcount
cd ~/wordcount
root@hp:~/wordcount# nano WCMapper.java
root@hp:~/wordcount# nano WCReducer.java
root@hp:~/wordcount# nano WCDriver.java
root@hp:~/wordcount# javac -classpath `hadoop classpath` -d . WCMapper.java WCReducer.java WCDriver.java
root@hp:~/wordcount# jar -cvf wordcount.jar *
added manifest
adding: WCDriver.class(in = 1838) (out= 937)(deflated 49%)
adding: WCDriver.java(in = 1448) (out= 502)(deflated 65%)
adding: WCMapper.class(in = 1786) (out= 722)(deflated 59%)
adding: WCMapper.java(in = 879) (out= 336)(deflated 61%)
adding: WCReducer.class(in = 1529) (out= 605)(deflated 60%)
adding: WCReducer.java(in = 813) (out= 313)(deflated 61%)
root@hp:~/wordcount# mkdir ~/input
nano ~/input/sample.txt
root@hp:~/wordcount# hadoop jar wordcount.jar WCDriver ~/input/sample.txt ~/output
2026-04-22 18:06:25,439 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-22 18:06:25,517 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-22 18:06:25,517 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-22 18:06:25,531 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-22 18:06:25,603 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your
application with ToolRunner to remedy this.
2026-04-22 18:06:25,696 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-22 18:06:25,715 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-22 18:06:25,869 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local96919616_0001
2026-04-22 18:06:25,869 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-22 18:06:25,957 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-22 18:06:25,957 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-22 18:06:25,957 INFO mapreduce.Job: Running job: job_local96919616_0001
2026-04-22 18:06:25,958 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-22 18:06:25,962 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-22 18:06:25,962 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup fa
ilures: false
```

```
                Bytes Written=81
root@hp:~/wordcount# cat ~/output/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
root@hp:~/wordcount#
```